AI SERVICES TRAINING PROGRAMME · SPECIALIZATION

AI Full-Stack & Deployment Engineer

Delivery Plan · 10 Sessions × 2 Hours

2.A Deployment Engineering › 2.B Full-Stack › Capstone & Shark Tank

Track 2 is for participants who will take models from notebook to live production and build the user-facing applications around them. 2.A covers serving stacks, CI/CD, monitoring and cloud deployment. 2.B builds production AI applications — streaming UIs, multimodal experiences and RAG-backed product features. It runs as 2-hour sessions over 10 days.

01 SPECIALIZATION HIGHLIGHTS

Specialization	AI Full-Stack & Deployment Engineer
Focus	Taking models from notebook to live production and building the applications around them.
Structure	Two parts — 2.A Deployment Engineering (5 sessions) + 2.B Full-Stack (5 sessions)
Duration	20 hours · 10 sessions × 2 hours · delivered over 10 days
Cadence	2.A across 5 sessions, then 2.B across 5 sessions; a hands-on lab in every session
Delivery	Instructor-led, with applied hands-on in each session
Assessment	1-hour assessment after each part
Capstone	Group project (max 3) · 10–15 days · defended Shark-Tank style
Capstone domains	Insurance · Banking · Healthcare · Supply Chain · Logistics

02 DELIVERY PLAN — PART 2.A

2.A Deployment Engineering	10 HOURS · 5 SESSIONS

Serving stacks, CI/CD, monitoring and cloud deployment.

Every session runs the same rhythm — concept framing, an instructor-led walkthrough, a guided hands-on lab, and a short review — within 2 hours.

SESSION 1 Day 1 — 2 hours

Objective *Choose and benchmark model-serving stacks for AI workloads.*

Topics Covered	Hands-On Lab (Applied)
• **Model serving stacks** – KServe, BentoML, vLLM, SGLang, NVIDIA Triton – deployment patterns on Kubernetes	• Package the same model with FastAPI, BentoML and vLLM; benchmark cold start, p95 latency and tokens/sec; pick a default with rationale

SESSION FLOW Concept framing & walkthrough ~45 min · Guided hands-on lab ~60 min · Review & wrap-up ~15 min

SESSION 2 Day 2 — 2 hours

Objective *Apply container-packaging discipline and deploy to managed cloud runtimes.*

Topics Covered	Hands-On Lab (Applied)
• **Container packaging discipline** – base images, GPU drivers, image size, supply-chain hygiene • **Cloud deployment targets** – AWS App Runner, Azure Container Apps, Google Cloud Run – managed Kubernetes (AKS, GKE, EKS) – cost & operability trade-offs	• Deploy the model to a managed cloud runtime with autoscaling, secrets, IAM and health probes

SESSION FLOW Concept framing & walkthrough ~45 min · Guided hands-on lab ~60 min · Review & wrap-up ~15 min

SESSION 3 Day 3 — 2 hours

Objective *Gate releases with CI/CD, canary rollouts and rollback.*

Topics Covered	Hands-On Lab (Applied)
• **CI/CD for AI** – GitHub Actions with Promptfoo eval gates – canary, blue-green and rollback	• Build a CI/CD pipeline (GitHub Actions) with Promptfoo eval gates; demonstrate a blocked merge on regression • Implement a canary deployment with automated traffic shift on live eval scores; verify rollback

SESSION FLOW Concept framing & walkthrough ~45 min · Guided hands-on lab ~60 min · Review & wrap-up ~15 min

SESSION 4 Day 4 — 2 hours

Objective *Instrument production monitoring and regulator-grade audit logging.*

Topics Covered	Hands-On Lab (Applied)
• **Production monitoring** – drift signals with Evidently, latency SLOs with Prometheus + Grafana, error budgets • **LLM audit logs for regulated industries** – what to retain, redaction at log time vs query time	• Stand up production monitoring (Prometheus + Grafana + Evidently) with drift signals; add audit-log retention with PII redaction
SESSION FLOW Concept framing & walkthrough ~45 min · Guided hands-on lab ~60 min · Review & wrap-up ~15 min	

SESSION 5 Day 5 — **2 hours**

Objective *Automate safe model upgrades and run a disciplined incident response.*

Topics Covered	Hands-On Lab (Applied)
• **Automated retraining triggers; shadow deployments for safe model upgrades** • **Incident response for AI services** – runbooks, escalation, rollback, postmortem discipline	• Run a shadow deployment of a new model version; compare traces in Langfuse against production; produce a go / no-go recommendation
SESSION FLOW Concept framing & walkthrough ~45 min · Guided hands-on lab ~60 min · Review & wrap-up ~15 min	

03 DELIVERY PLAN — PART 2.B

2.B Full-Stack — 10 HOURS · 5 SESSIONS

Production AI applications — streaming UIs, multimodal experiences and RAG-backed product features.

Every session runs the same rhythm — concept framing, an instructor-led walkthrough, a guided hands-on lab, and a short review — within 2 hours.

SESSION 1 Day 6 — **2 hours**

Objective *Build production AI APIs with FastAPI and pick the right application framework.*

Topics Covered	Hands-On Lab (Applied)
• **Production AI APIs with FastAPI** – streaming responses, async, Server-Sent Events, retries, idempotency • **Application frameworks** – Streamlit for rapid analytics delivery – Next.js + Vercel AI SDK for production user-facing apps	• Build a production FastAPI backend with streaming, async, Server-Sent Events and Pydantic structured outputs • Build a chat experience in Streamlit for rapid analytics-team delivery; compare trade-offs
SESSION FLOW Concept framing & walkthrough ~45 min · Guided hands-on lab ~60 min · Review & wrap-up ~15 min	

SESSION 2 Day 7 — **2 hours**

Objective *Abstract providers with LiteLLM and ship a streaming production UI.*

Topics Covered	Hands-On Lab (Applied)
• **Provider abstraction with LiteLLM** – routing across Bedrock, Azure OpenAI and Vertex AI with fallbacks and cost-based selection	• Build a streaming chat UI with Vercel AI SDK and Next.js 15 calling a model via AWS Bedrock / Azure OpenAI / Vertex AI • Add LiteLLM provider abstraction with cross-cloud routing fallbacks and cost-based selection
SESSION FLOW Concept framing & walkthrough ~45 min · Guided hands-on lab ~60 min · Review & wrap-up ~15 min	

SESSION 3 Day 8 — 2 hours

Objective *Add multimodal input and RAG-backed, cited application features.*

Topics Covered	Hands-On Lab (Applied)
• **Multimodal experiences** – image + text + document input with Claude vision, GPT-4o, Gemini 2.5 • **RAG-backed features inside apps** – retrieval, grounding, inline citation rendering, confidence surfacing	• Build a RAG-backed feature with inline citation rendering and confidence surfacing in the UI
SESSION FLOW Concept framing & walkthrough ~45 min · Guided hands-on lab ~60 min · Review & wrap-up ~15 min	

SESSION 4 Day 9 — 2 hours

Objective *Integrate single-agent tool-use via MCP inside product features.*

Topics Covered	Hands-On Lab (Applied)
• **Single-agent tool-use inside product features** – search, summarise, draft flows with MCP tool integration	• Add multimodal input (image + text) and integrate MCP tools for a single-agent product feature
SESSION FLOW Concept framing & walkthrough ~45 min · Guided hands-on lab ~60 min · Review & wrap-up ~15 min	

SESSION 5 Day 10 — 2 hours

Objective *Operate LLMOps and Responsible-AI controls inside the application layer.*

Topics Covered	Hands-On Lab (Applied)
• **LLMOps inside the app layer** – prompt versioning, Langfuse tracing, A/B testing prompts in production • **Responsible AI in code** – Guardrails AI content filters, refusal handling, PII redaction at the boundary	• Add prompt versioning and Langfuse tracing to the app; apply Guardrails AI content filters and PII redaction at the boundary
SESSION FLOW Concept framing & walkthrough ~45 min · Guided hands-on lab ~60 min · Review & wrap-up ~15 min	

04 SUGGESTED CAPSTONE PROJECTS

Ten suggested capstones spanning Insurance, Banking, Healthcare, Supply Chain and Logistics — each scoped for a group of up to three over 10–15 days on synthetic data. Every brief states the business problem, what to build, and the success metrics it will be judged on.

1 INSURANCE Claims Status Copilot

Business problem Call centres are flooded with "where's my claim?" queries that tie up agents.

What to build FastAPI + Next.js / Streamlit app answering claim status via RAG with citations; deploy to a managed cloud runtime with monitoring, audit logs and CI/CD eval gates.

Success metrics Answer accuracy, p95 latency, SLO / uptime adherence, simulated deflection rate, cost per query.

2 INSURANCE Policy Comparison Web App

Business problem Comparing policy documents by hand is slow and error-prone.

What to build Multimodal app (upload policy PDFs) with LiteLLM routing across providers, grounded comparison output, Guardrails and PII redaction; containerized and deployed.

Success metrics Comparison accuracy, p95 latency, provider-failover success rate, guardrail catch rate.

3 BANKING Loan Eligibility Assistant

Business problem Pre-qualification takes staff time and yields inconsistent answers.

What to build Streaming chat app with RAG over eligibility rules and a FastAPI backend; CI/CD with Promptfoo eval gates; audit logs.

Success metrics Eligibility-answer accuracy, faithfulness, blocked-merge rate on regressions, latency.

4 BANKING Fraud Triage Dashboard

Business problem Analysts triage fraud alerts without a fast, explainable interface.

What to build Deploy a fraud-scoring model behind FastAPI; Streamlit ops dashboard with explanations; Prometheus / Grafana SLOs and Evidently drift.

Success metrics p95 latency, throughput, drift-alert accuracy, SLO adherence.

5 HEALTHCARE Patient Intake Copilot

Business problem Manual intake (forms + documents) is slow and error-prone, and PII must be protected.

What to build Multimodal intake app (form + document upload) with PII redaction at the boundary; deploy HIPAA-aligned with audit-log retention.

Success metrics Extraction accuracy, PII-redaction recall, simulated intake-time reduction, audit completeness.

6 HEALTHCARE Readmission Risk Service

Business problem A completed readmission model sits in a notebook; operations cannot consume it.

What to build Package (FastAPI + Docker) and deploy to a managed runtime with autoscaling and health probes; monitoring, CI/CD eval gates and rollback.

Success metrics p95 latency, cold-start time, SLO adherence, rollback success, audit-log coverage.

7 SUPPLY CHAIN Supplier Onboarding Portal

Business problem Onboarding suppliers requires repetitive document handling and Q&A.

What to build Full-stack app with document upload, RAG-backed Q&A and MCP tool-use (mock ERP); deployed with CI/CD.

Success metrics Q&A accuracy, task-completion rate, latency, deployment eval-gate pass rate.

8 SUPPLY CHAIN Demand Forecast Serving API

Business problem Forecasting models are not served reliably for planning systems to consume.

What to build Serve a forecasting model (vLLM / BentoML-style) with canary rollout, Grafana SLOs and a shadow deployment for the next version.

Success metrics p95 latency, throughput, canary rollback correctness, SLO adherence.

9 LOGISTICS Delivery Exception Assistant

Business problem Ops need fast, cited resolutions for delivery exceptions.

What to build Streaming app surfacing exception resolutions with citations and confidence; deploy with observability (Langfuse, Prometheus / Grafana).

Success metrics Resolution accuracy, faithfulness, p95 latency, uptime.

10 LOGISTICS Route Scoring Model Service

Business problem A route-scoring model needs a safe, observable production rollout.

What to build Deploy behind FastAPI with monitoring, audit logs and a shadow deployment for the next version; produce a go / no-go recommendation from traces.

Success metrics p95 latency, drift signals, shadow-vs-prod delta, rollback success.

05 CAPSTONE GUIDELINES & SHARK TANK PRESENTATION

Each group takes one capstone to a defended Shark-Tank presentation. The capstone converts the specialization's learning into a demonstrated, review-tested system.

Team & Execution

- Groups of up to 3 participants.
- Timeline: 10–15 days from assignment to the Shark Tank defence.
- Use synthetic or public data only — no real customer data, PII or PHI. Generate realistic, real-format test data.
- Pick one domain per project — Insurance, Banking, Healthcare, Supply Chain or Logistics.
- The build must integrate and demonstrate the core competencies of this specialization.

What “done” means

- It runs — a working end-to-end system, not slideware or a notebook of fragments.
- It's measured — report real numbers against the project's success metrics (accuracy, latency, cost, quality).
- It's safe — at least one evaluation or safety component (guardrails, refusal, hallucination or drift check).
- It's reproducible — a code repository with a README that lets the panel run it.

Pre-Pitch Deliverables

- Code repository with a README — setup, run instructions and an architecture diagram.
- Evaluation report with the metrics named in the project brief.
- A working demo — live, or a recorded fallback in case of network failure.
- A 20-minute pitch deck and a one-page summary of architecture, trade-offs and limitations.

SHARK TANK PRESENTATION — 30 MINUTES

20 minutes presentation + 10 minutes Q&A. Present the working system and walk its architecture, trade-offs, evaluation evidence, cost and limitations. The panel challenges the team live; the team defends its decisions.

20-Minute Presentation — Suggested Split

Time	Segment
3 min	**Problem & business case** — the pain, who feels it, and the cost of the status quo.
4 min	**Solution architecture** — the pipeline / agents and which specialization competencies were integrated, and why.
7 min	**Live demo** — show it working on real-format (synthetic) inputs.
4 min	**Results & evaluation** — metrics against the baseline — quality, latency, cost.
2 min	**Limitations & next steps** — what it does not do yet, and what would come next.
10 min	**Panel Q&A** — the panel challenges architecture, trade-offs, evaluation and cost; every member defends their area.

Evaluation Rubric

Criterion	Weight	What is assessed
Functionality	**25%**	The system works end-to-end and does what it claims.

Criterion	Weight	What is assessed
Technical Depth	**20%**	Sound engineering and correct use of the specialization's tools and patterns.
Evaluation & Safety	**15%**	Rigorous evaluation, guardrails, and honest handling of limitations.
Usefulness	**15%**	Clear real-world applicability and business relevance.
Novelty	**15%**	Originality of the approach and problem framing.
Presentation	**10%**	Clarity of the pitch and quality of the defence under Q&A.